Physics 610
Fall 2023
S. Khlebnikov

Homework 5

Due Wednesday, October 18

1. The Lagrangian of a relativistic particle (mass m, charge q) moving in an external electromagnetic field is
$$L = -mc^2 \left(1 - \frac{v^2}{c^2}\right)^{1/2} + \frac{q}{c}\mathbf{v}\cdot\mathbf{A}(\mathbf{r},t) - q\Phi(\mathbf{r},t)\,,$$
where $\mathbf{v} = d\mathbf{r}/dt$ is the particle's velocity, $\mathbf{A}$ and Φ are the vector and scalar potential at the particle's position, and c is the speed of light. (This expression uses the Gaussian units.)
(a) Compute the canonical momentum conjugate to $\mathbf{r}$ and derive the Hamiltonian. As a check of your calculation, compare your result to Eq. (8.56) of the textbook.
(b) Verify that Hamilton's equations lead to the standard expression for the Lorentz force, including both electric and magnetic components.

2. For a system of two particles described by the Hamiltonian
$$H = \frac{\mathbf{p}_1^2}{2m_1} + \frac{\mathbf{p}_2^2}{2m_2} + V(\mathbf{R}_{rel})\,,$$
where $V(\mathbf{R}_{rel})$ is a potential that depends only on the relative coordinate
$$\mathbf{R}_{rel} = \mathbf{r}_1 - \mathbf{r}_2\,,$$
a convenient canonical transformation is to the relative and center-of-mass coordinates and momenta:
$$(\mathbf{r}_1, \mathbf{r}_2, \mathbf{p}_1, \mathbf{p}_2) \to (\mathbf{R}_{rel}, \mathbf{R}_{cm}, \mathbf{P}_{rel}, \mathbf{P}_{cm})\,.$$
There are six generalized coordinates in each set, so let's denote the set on the left as (q_i, p_i) and the set on the right as (Q_i, P_i), where $i = 1, \ldots, 6$.
(a) Expression for $\mathbf{R}_{rel}$ is given above. Write down expressions for other variables from the second set. As a check that the transformation is canonical, verify that it preserves Poisson brackets between coordinates and momenta.
(b) For any canonical transformation for which the matrix $\partial P_i/\partial p_j$ (at fixed q and fixed time t) is nondegenerate, one can define a generating function $F_2(q, P, t)$ (where q stands for all q_i, and P for all P_i) such that
$$p_i = \frac{\partial F_2}{\partial q_i}\,, \qquad Q_i = \frac{\partial F_2}{\partial P_i}\,. \tag{1}$$

In the present case, one can look for F_2 in the form

$$F_2(q, P) = \boldsymbol{f}_1(q) \cdot \mathbf{P}_{rel} + \boldsymbol{f}_2(q) \cdot \mathbf{P}_{cm}\,,$$

where $\boldsymbol{f}_1(q)$, $\boldsymbol{f}_2(q)$ are 3-component vector functions. Find these functions. Please make sure that you have verified all of the relations (1).

(c) Find the new Hamiltonian $K(P, Q)$ and confirm that it separates into a sum of Hamiltonians for the center of mass and relative motions.